

Problem Sheet 6

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1. Construct an automata which can accept the following languages:

- The set of words over $\{0, 1\}$ which contain an even number of 0's and least one 1.
- The set of words over $\{0, 1\}$ such that some pair of 0's is separated by a string of length $4i$, $i \geq 0$.
- The set of words over $\{0, 1, 2\}$ which are base 3 representations of odd numbers.
- Let $\Sigma = \left\{ \begin{pmatrix} a \\ b \\ c \end{pmatrix} \mid a, b, c \in \{0, 1\} \right\}$. A string over Σ gives three rows of 0's and 1's. Treat each row as a binary number.
The set of words $\{w \in \Sigma^* \mid \text{the top row and the middle row sum to the bottom row}\}$.

2. Which of the following languages are regular? Which are FO definable?

- The set of words over $\{a, b\}$ which has equal number of occurrences of ab and ba . For example, aba is in the language, while $abab$ is not.
- The set of words over $\{a, b\}$ which do not have any 2 consecutive b's.
- The set of words over $\{0, 1\}$ with at most one pair of consecutive 0's and at most one pair of consecutive 1's.
- The set of strings over $\{0, 1\}$ such that the second symbol from both ends is 0.
- Let $\Sigma = \left\{ \begin{pmatrix} a \\ b \end{pmatrix} \mid a, b \in \{0, 1\} \right\}$. A string over Σ gives two rows of 0's and 1's. Treat each row as a binary number. The set of words $\{w \in \Sigma^* \mid \text{the top row is larger than the bottom row}\}$

3. Consider the following FO formulae. In each case, answer the following questions:

- What is $L(\varphi)$?
- What is $\overline{L(\varphi)}$?
- Is $L(\varphi)$ regular?
- Is $\overline{L(\varphi)}$ regular?

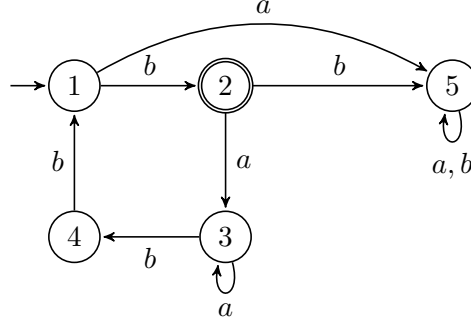
(a) $\forall x(x \neq x)$

(b) $\exists x \exists y [x < y \wedge Q_b(x) \wedge Q_a(y) \wedge \forall z [(x < z < y) \rightarrow Q_a(z)]]$

(c) $\exists x[Q_a(x) \wedge \exists y[S(x, y) \wedge \forall z[z \leq y]]]$

(d) $\exists x \forall y[x \leq y \wedge Q_a(x)] \wedge \exists x \forall y[y \leq x \wedge Q_b(x)] \wedge$
 $\forall x \forall y[Q_a(x) \wedge S(x, y) \rightarrow Q_b(y)] \wedge \forall x \forall y[Q_b(x) \wedge S(x, y) \rightarrow Q_a(y)]$

4. Consider the following automaton. What is the language L accepted? Can you write an FO formula φ such that $L = L(\varphi)$?



5. (a) Let A be a DFA accepting the language L . Is the reverse of all the strings accepted by $L(A)$ regular?
- (b) Let L be a regular language over $\{a, b, c\}$. Define the projection of L with respect to $\{b, c\}$ denoted $L \downarrow \{b, c\}$ as the language

$$\{w' \mid w' \text{ is obtained from } w \in L \text{ after deleting all occurrences of symbol } a\}$$

Is $L \downarrow \{b, c\}$ regular?

- (c) Show that every NFA can be converted into an equivalent one with a single accepting state.
- (d) Let $N = (Q, \Sigma, \delta, q_0, F)$ be a DFA. Construct an automaton $N_1 = (Q, \Sigma, \delta_1, q_0, F_1)$ as follows:
- $F_1 = F \cup \{q_0\}$
 - Define δ_1 such that for any $q \in Q$ and $a \in \Sigma \cup \{\epsilon\}$,

$$\delta_1(q, a) = \delta(q, a) \text{ for } q \notin F \text{ or } a \neq \epsilon$$

$$\delta_1(q, a) = \delta(q, a) \cup \{q_0\} \text{ for } q \in F \text{ and } a = \epsilon$$

Is $L(N_1) = (L(N))^*$?

- (e) Let L be a regular language. Consider the language L' defined as

$$\{w \in L \mid \text{no proper prefix of } w \text{ is in } L\}$$

Show that L' is regular.

6. Let $L \subseteq \{a, b\}^*$ be the language consisting of all palindromes: that is, strings like abba that are the same backwards and forwards. Using the pigeonhole principle, show that L is not regular.

7. Recall that we defined an angelic acceptance condition for NFAs in class : a word w is accepted whenever it has atleast one accepting run. Under this, we showed that the languages accepted by NFAs are regular. Consider the following *devilish* acceptance condition, which says that an NFA M accepts a word x iff every possible computation of M on x ends in an accept state. Show that NFAs with the devilish acceptance condition recognize the class of regular languages.